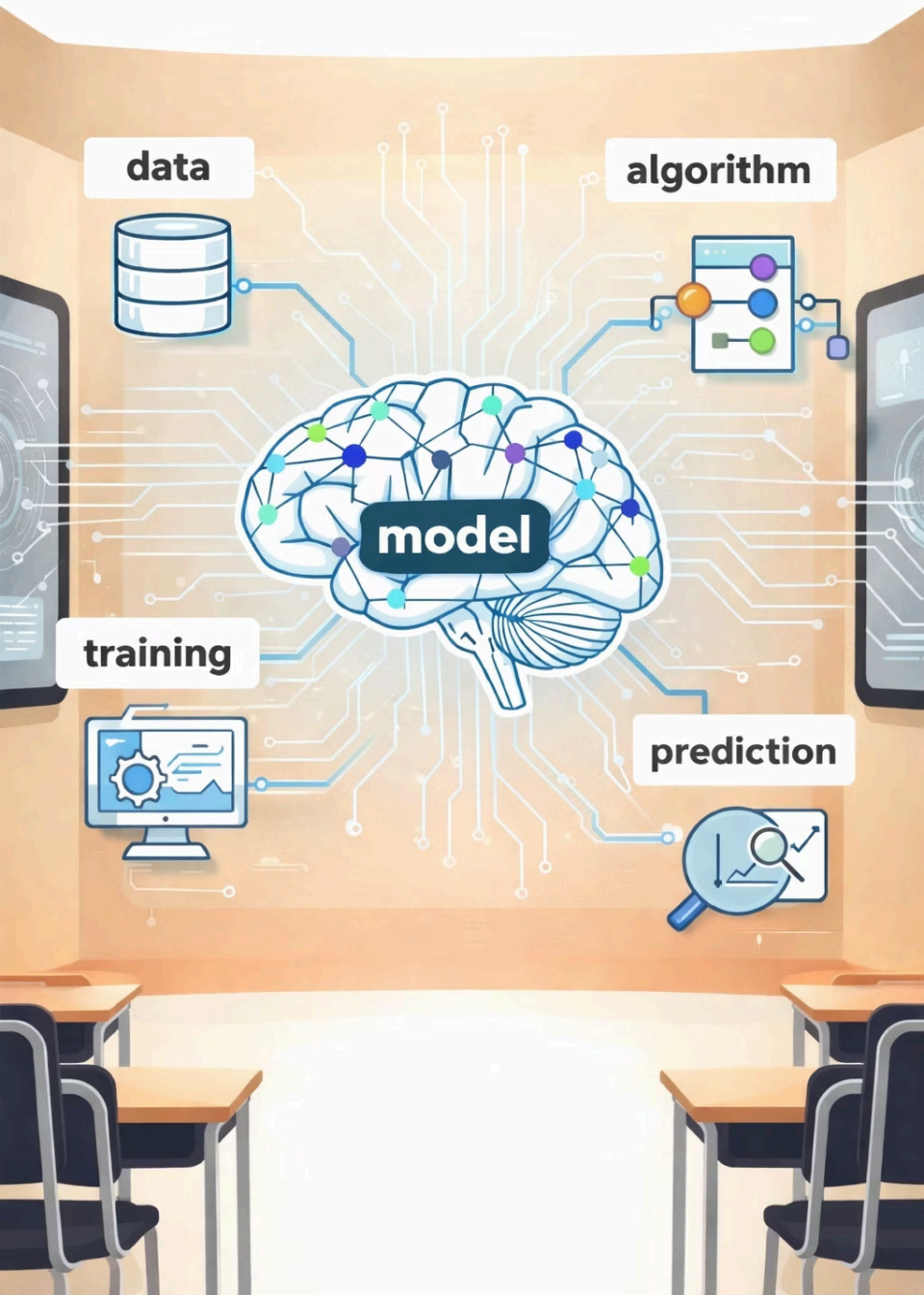




Session 2: AI Concepts & Terminology

Last session we learned what AI *does*. Today we're going to learn the **special words** people use to talk about AI.

Think of it like learning the rules of a new game - once you know the words, everything else gets way easier.

Objectives

By the end of this session, you will be able to:

01

Data

Explain what **data** is - the stuff AI learns from

02

Algorithm

Explain what an **algorithm** is - the recipe AI follows-

03

Model

Explain what a **model** is - what AI becomes after learning

04

Prediction

Explain what a **prediction** is - AI's best guess

① Put it all together: **Data + Algorithm → Training → Model → Prediction**

Recap & Homework Share

Welcome back! Let's start with your story.

Show Me Your Storybook!

- What was the AI good at?
- Did it surprise you with anything?
- Did it ever say something silly or weird?

From Last Time, Do You Remember?

- What does AI mean?
- What's the Sense → Think → Act pattern?
- What's the big rule? (AI is a tool - YOU are the boss!)



Warm-Up - Words You've Heard

Before we dive in, let's see what you already know. Think about these questions:



"Algorithm"

Have you ever heard this word before? Where did you hear it?



"Training"

What do you THINK "training" means for a computer?

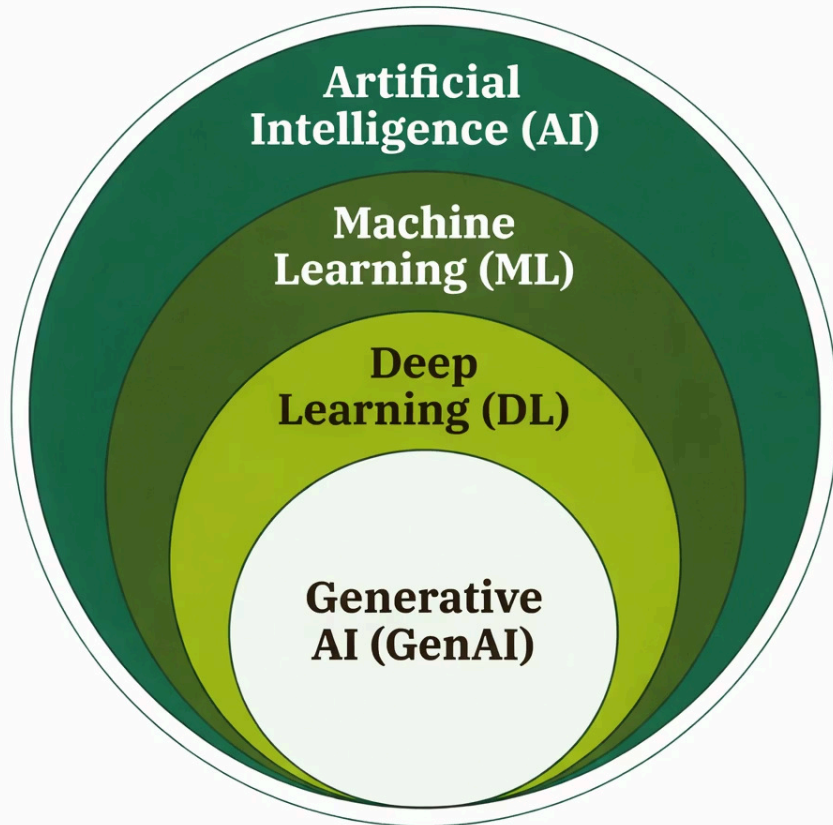


Your Questions

What questions do you have about how AI actually works inside?

The Big Picture - Where Today's Words Fit

Before we learn the words, let's see where they fit in the AI world.



Artificial Intelligence

Machines doing **smart things** - broad and big

Machine Learning

AI that **learns from examples** instead of fixed rules

Deep Learning (DL)

A powerful type of ML using brain-like networks

Generative AI (GenAI)

AI that **CREATES** new things - text, images, music, code

Words like *data*, *algorithm*, *training*, *model*, *prediction* - these show up everywhere in AI.



Remember your Storybook from last time? It was an AI that **LEARNED** how to write stories from millions of examples. That's Machine Learning - and we'll learn all about it next session!

DATA

Data = Information.

It's the **stuff** AI learns from. Without data, AI has nothing to work with.



**Chef with No
Ingredients** 🍲

Can't cook anything
without ingredients



**Painter with No
Paint** 🎨

Can't create art
without paint



Car with No Fuel ⛽

Can't go anywhere
without fuel

📄 **Data is the fuel of AI**



What Kinds of Data Does AI Use?

AI can learn from lots of different stuff:



Numbers

Temperatures, scores, prices



Words

Books, texts, your messages



Pictures

Photos, drawings, selfies



Sounds

Songs, voices, podcasts



Videos

TikToks, YouTube clips



Sensors

Heartbeat on a smartwatch, GPS location, motion

 Every time you watch, like, type, or take a photo - **you're making data.**

Mini Activity 1 - Data Detective

Look at this list. What kind of data is each one? Choose: **Numbers, Words, Pictures, Sounds, Videos, or Sensors**

#	Item	Kind of Data
1	A selfie on your phone	
2	A text message saying "Hi!"	
3	A voice memo of you singing	
4	The temperature outside (72°F)	
5	A TikTok dance video	
6	Your high score in a game	
7	Your heartbeat tracked by a smartwatch	
8	Your phone's GPS showing where you are	

Mini Activity 1 - Answers

#	Item	Kind of Data
1	A selfie on your phone	Pictures
2	A text message saying "Hi!"	Words
3	A voice memo of you singing	Sounds
4	The temperature outside (72°F)	Numbers
5	A TikTok dance video	Videos
6	Your high score in a game	Numbers
7	Your heartbeat tracked by a smartwatch	Sensors
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The Most Important Data Rule

"Garbage In = Garbage Out"

If the data is bad, the AI will be bad. No matter how smart it is.

Only Brown Dogs

Train an AI only with photos of brown dogs → it won't recognize white dogs or black dogs

Only Sunny Days

Train an AI only on sunny day photos → it can't recognize streets when it's raining or snowing

Only Old Books

Train an AI to translate using only fancy old books → it'll talk like a robot from the 1800s

✓ **Good data → Smart AI.**

Bad data → Bad AI.

This is why people who build AI have to think really carefully about WHAT data they use.

ALGORITHM

Algorithm = a step-by-step set of instructions.

You already use algorithms every day - you just don't call them that!

Algorithm for Brushing Your Teeth

1. Pick up toothbrush
2. Put toothpaste on it
3. Brush for 2 minutes
4. Rinse mouth
5. Put toothbrush back

Algorithm for Getting Ready for School

1. Wake up
2. Get dressed
3. Eat breakfast
4. Pack bag
5. Leave



Recipe = Algorithm

Here's the connection - a recipe and an algorithm are the same idea in different worlds:

Recipe (for a cook)	Algorithm (for a computer)
Uses ingredients	Uses data
Step-by-step instructions	Step-by-step instructions
Makes a dish	Makes a result
Any cook can follow it	Any computer can run it

Mini Activity - Algorithm Hunt

You follow algorithms every day without even realizing it! Pick **THREE** things you do step-by-step every day and write down YOUR algorithm for each one.

Need Ideas? Try One of These:

- How you brush your teeth 
- How you make your favorite sandwich 
- How you log into your favorite game 
- How you get ready for bed 
- How you pack your school bag 
- How you tie your shoes 

Example - Algorithm for Making Toast:

1. Get bread from the kitchen
2. Put one slice in the toaster
3. Push the lever down
4. Wait until it pops up
5. Spread butter on top
6. Eat 

☑ If a computer could follow your steps, you just wrote an algorithm!



TRAINING

Training = how AI learns from data.

It's like studying for a really, really big test - except the AI studies **MILLIONS** of examples.

Training is the **process**. The model is the **result**. These two ideas work together to create every AI system you've ever used.

How Training Works (Simply)



Data feeds into the algorithm, which goes through training, and produces a model.



The Flashcard Story

Imagine you've never seen a dog before. How would you learn?

1

📇 Flashcards

Each one has a picture and says "Dog" or "Not a dog" - *This is your DATA*

2

📋 Study Method

Look, guess, check, repeat - *This is your ALGORITHM*

3

📖 Practice

You practice with thousands of flashcards - *This is TRAINING*

4

🧠 Expert!

Now you can look at ANY new photo and say "Dog!" or "Not a dog!" - *You are now a trained MODEL 🎉*

📌 That's exactly how AI learns. Just with way more flashcards.

Mini Activity 2 - Spot the Step

For each one, tell me: Is it **DATA**, **ALGORITHM**, **TRAINING**, or **MODEL**?

A folder of 5,000 pictures of cats and dogs

Answer: **DATA**

"Look at picture, guess cat or dog, check answer, repeat"

Answer: **ALGORITHM**

The AI looks at all 5,000 pictures over and over, getting better each time

Answer: **TRAINING**

The finished AI that can now tell cats from dogs

Answer: **MODEL**

MODEL

A model is the AI AFTER it's done training.

It's the **finished, ready-to-use** version.

What Goes IN	What Comes OUT
 Algorithm (the recipe)	 Model (the trained expert)
 Data (the examples)	Knows patterns from the data
 Training (the practice)	Ready to make predictions

Examples of Models You Already Use:

 ChatGPT

A model trained on billions of web pages

 Face Unlock

A model trained on photos of YOUR face

 Netflix

A model trained on what millions of people watched

Algorithm vs. Model

These two words trip people up all the time. Here's the easy way to remember:

Algorithm

- The **recipe**
- The **workout plan**
- Blank sheet music 🎵
- General instructions
- **BEFORE** learning

Model

- The **finished dish**
- The **trained athlete**
- A musician who knows the song 🎻
- Specific skill
- **AFTER** learning

❶ **Algorithm = BEFORE learning. Model = AFTER learning.**

A million different people can follow the same recipe - but they each become their own chef. Same with AI: the same algorithm + different data = different models.



PREDICTION

Prediction = AI's best guess about something new.

After training, when the AI sees something it's never seen before, it guesses.

Look at all the predictions AI makes for you every day:



Netflix

Predicts what show
you'll like next



Google Maps

Predicts how long
your trip will take



Your Phone

Predicts the next
word as you type



Weather App

Predicts if it'll rain
tomorrow



Amazon

Predicts what you
might want to buy



Spotify

Predicts the next
song you'll love

The AI Recipe - All Together Now

Let's put all five words together. This is the AI Recipe:



- ✓ This recipe works for **EVERY AI system in the world**. YouTube. ChatGPT. Snapchat. Self-driving cars. All of them.

Example - How YouTube Uses the Recipe

Let's trace it through one app you definitely use:



Data - The Fuel of AI

What you watch, like, skip, search for



Algorithm

YouTube's instructions for finding patterns



Training

The algorithm studies your viewing data + millions of other people's



Model

A personalized YouTube model - one just for you



Prediction

"I bet you'll want to watch THIS next"
→ next video suggestion

 Now you know exactly how YouTube does it.



Hands-On Activity - Explore Gemini

Now let's actually test these ideas. We're going to use **Google Gemini**.

Gemini is a **model** that was **trained** on billions of pages of text. Let's see if we can spot the recipe in action.

Try These Together with Your Mentor:

1

Ask About Training

Ask Gemini: *"What were you trained on?"*

2

Change the Audience

Ask: *"Explain how a refrigerator works like I'm 5 years old."* Then ask: *"Now explain it like I'm in high school."*

3

Finish the Story

Type: *"Once upon a time, in a galaxy far, far away, ___"* and let Gemini finish it

ⓘ Watch what happens. You just saw: the **MODEL** (Gemini) answering you, **TRAINING** data behind the answers, and **PREDICTIONS** (every word it says is a prediction!)

What Did You Notice?

Discuss:

Was It Really Thinking?

When Gemini answered, was it really *thinking*? Or was it making predictions based on patterns?

What Surprised You?

Did anything Gemini said catch you off guard? What was unexpected?

Did It Get Anything Wrong?

Did it ever say something that wasn't quite right? What happened?



Remember: Every word Gemini says is a **prediction**. It's not thinking. It's pattern-matching.

What We Learned this Session:

You now speak the language of AI.

Data

The examples AI learns from - and
"garbage in = garbage out"

Algorithm

The step-by-step recipe AI follows

Training

How AI learns from data - like studying
flashcards

Model

The finished, trained AI ready to use

Prediction

AI's best guess on something new

The AI Recipe: Data + Algorithm → Training → Model → Prediction

Your Homework - AI Detective

Pick **ONE AI** you use a lot:

Youtube / Netflix

Spotify

Google Maps

Snapchat Filters

Chatgpt / Gemini

Now answer these 5 questions about it:

1 Data

What kind of **data** do you think it was trained on?

2 Algorithm

What is its **algorithm** trying to do?

3 Prediction

What does it **predict**?

4 Training

How does **training** make it better over time?

5 Model

What's the **model** doing when you use it?

 Write 1–2 sentences for each. Bring it next session.  About 15 minutes.

Next Time - Session 3: What is Machine Learning?

We talked about how AI **learns from data** - that's actually called **Machine Learning**.

Next session, we'll go deeper:



Why Rules Aren't Enough

Why can't computers just follow fixed rules?



Finding Patterns

How do they find patterns in data?



Live Demo!

We'll watch a real ML model learn LIVE using **Google Teachable Machine!**



Bring your AI Detective notes! See you next session.